

Data Analytics Final Project

Product Sales & Region Analysis

Cleaning & preparation, exploratory analysis, dashboard and business insights
1,500 orders | Jan 2023 to Jun 2025

\$4.38M

Net sales

1,500

Orders

15,616

Units

\$2,920

Avg order

24.8%

Return rate

6.0 d

Avg delivery

Contents

1. Problem statement
2. Dataset information
3. Data cleaning and preparation
4. Exploratory data analysis (charts and matrices)
5. Dashboard (Power BI)
6. Key business insights and recommendations
7. Limitations

1. Problem statement

The business sells seven products through five regions, four stores and six salespeople, but has no consolidated view of what drives revenue and where it leaks. This project cleans the raw order data, analyses it, and summarises it in dashboard-style charts to answer:

- Which regions, products and salespeople generate the most net sales, and how has revenue moved over time?
- How much revenue is given away through discounts and promotions?
- What is the return rate, and does it vary by product, region or discount depth?
- Are there unusual orders or delivery issues that need operational review?

2. Dataset information

Source: **Product-Sales-Region-Raw-Data.xlsx**, one sheet, **1,500 rows x 19 columns**. Each row is one order.

Group	Columns
Order	OrderID, Date, OrderDate, DeliveryDate, CustomerName, CustomerType (Retail / Wholesale)
Geography and people	Region (5), StoreLocation (4), RegionManager, Salesperson (6)
Product and price	Product (7), Quantity, UnitPrice, Discount (0 / 5 / 10 / 15%), TotalPrice, ShippingCost
Commercial	PaymentMethod (5), Promotion (FREESHIP, SAVE10, WINTER15, blank), Returned (0/1)

The data has no cost or profit field, so all findings describe sales rather than margin.

3. Data cleaning and preparation

The raw and cleaned workbooks were compared and every step was re-verified in code. The cleaned file is **Product-Sales-Region-cleaned-Data.xlsx** (sheet Sales, 1,500 rows x 22 columns, no blanks).

Step	Count	Note
Rows loaded	1,500	Raw file
Duplicate rows removed	0	Full-row duplicates
Duplicate OrderIDs removed	0	Each row is one order
Blank Promotion -> 'No Promotion'	370	Kept so these orders appear in analysis
Column dropped: OrderDate	1	Identical to Date on every row
Columns added	4	DeliveryDays, GrossSales, NetSales, ReturnedFlag
NetSales vs TotalPrice mismatches	0	0 means derivations reconcile
Rows after cleaning	1,500	22 columns

Derived fields

Field	Definition
DeliveryDays	DeliveryDate minus Date (range 2 to 10 days)
GrossSales	Quantity x UnitPrice
NetSales	GrossSales x (1 - Discount); reconciles with TotalPrice on all 1,500 rows
ReturnedFlag	Kept / Returned label from the 0/1 Returned field

4. Exploratory data analysis

Methods: descriptive statistics, grouped summaries, time-series aggregation, Pearson correlation and 1.5xIQR outlier screening.

Metric	Value
Gross sales	\$4,727,892.94
Net sales	\$4,379,992.43
Discounts given	\$347,900.51 (7.4% of gross)
Average / median net sales per order	\$2,919.99 / \$2,174.72
Return rate	24.8% (372 orders)
Average delivery	6.04 days
High-value outliers (above IQR fence)	17 orders above \$9,736.10

4.1 Trend over time

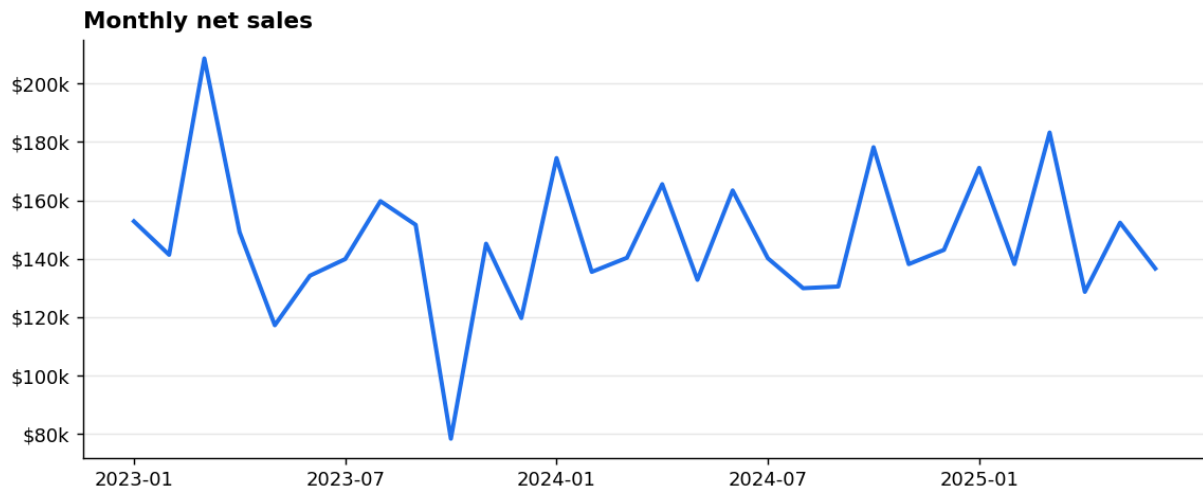
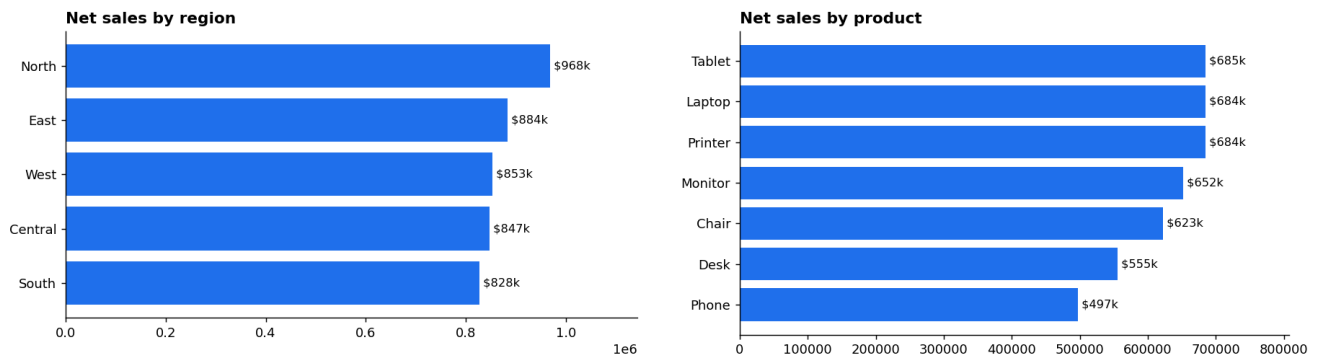


Figure 1. Monthly net sales. Peak 2023-03 (\$208,549); low 2023-10 (\$78,446).

Year	Orders	Net sales	Avg order
2023	579	\$1,697,870	\$2,932
2024	584	\$1,771,955	\$3,034
2025	337	\$910,168	\$2,701

2025 covers January to June only. First-half net sales: 2023: \$903,340, 2024: \$911,988, 2025: \$910,168.

4.2 Region, product and salesperson



Figures 2-3. Net sales by region and by product.

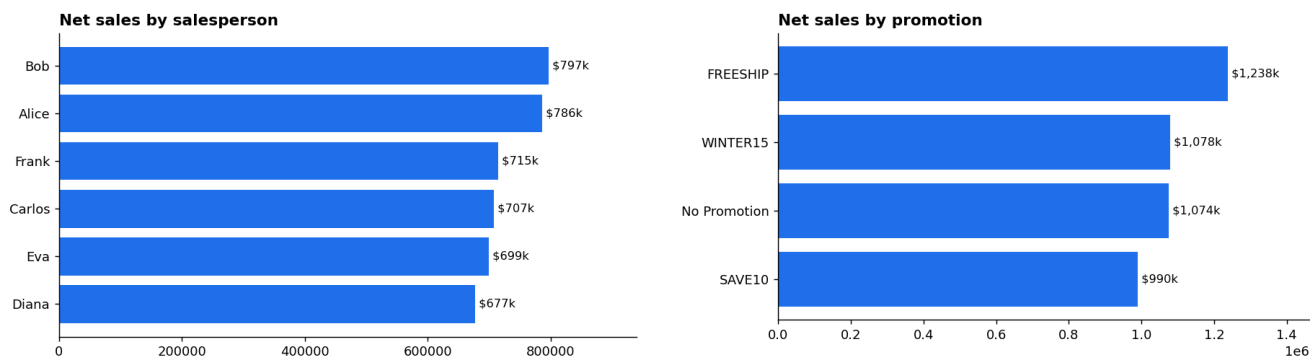
Regional detail

Region	Orders	Net sales	Units	Return rate	Avg delivery
North	309	\$967,958	3,442	22.7%	6.39 d
East	311	\$883,634	3,078	26.0%	6.00 d
West	284	\$853,479	3,041	26.1%	6.01 d
Central	301	\$847,154	3,102	23.3%	5.88 d
South	295	\$827,768	2,953	26.1%	5.93 d

Product detail

Product	Orders	Net sales	Units	Return rate	Avg delivery
Tablet	240	\$684,539	2,577	24.2%	6.13 d
Laptop	226	\$684,417	2,408	27.4%	6.03 d
Printer	211	\$684,387	2,336	22.3%	6.10 d
Monitor	212	\$651,629	2,177	26.4%	6.32 d
Chair	209	\$622,589	2,122	28.2%	5.86 d
Desk	207	\$555,267	2,026	21.7%	6.11 d
Phone	195	\$497,163	1,970	23.1%	5.70 d

Salesperson detail



Figures 4-5. Net sales by salesperson and by promotion.

Salesperson	Orders	Net sales	Units	Return rate	Avg delivery
Bob	243	\$796,781	2,661	26.3%	6.02 d
Alice	259	\$786,166	2,732	24.7%	5.94 d
Frank	243	\$714,642	2,524	22.6%	6.15 d
Carlos	270	\$707,167	2,699	28.1%	6.05 d
Eva	236	\$698,670	2,452	21.6%	6.15 d
Diana	249	\$676,568	2,548	24.9%	5.95 d

Promotion detail

Promotion	Orders	Net sales	Units	Return rate	Avg delivery
FREESHIP	419	\$1,238,077	4,397	23.9%	5.90 d
WINTER15	373	\$1,078,012	3,827	24.7%	6.11 d
No Promotion	370	\$1,074,247	3,787	24.9%	6.10 d
SAVE10	338	\$989,657	3,605	26.0%	6.09 d

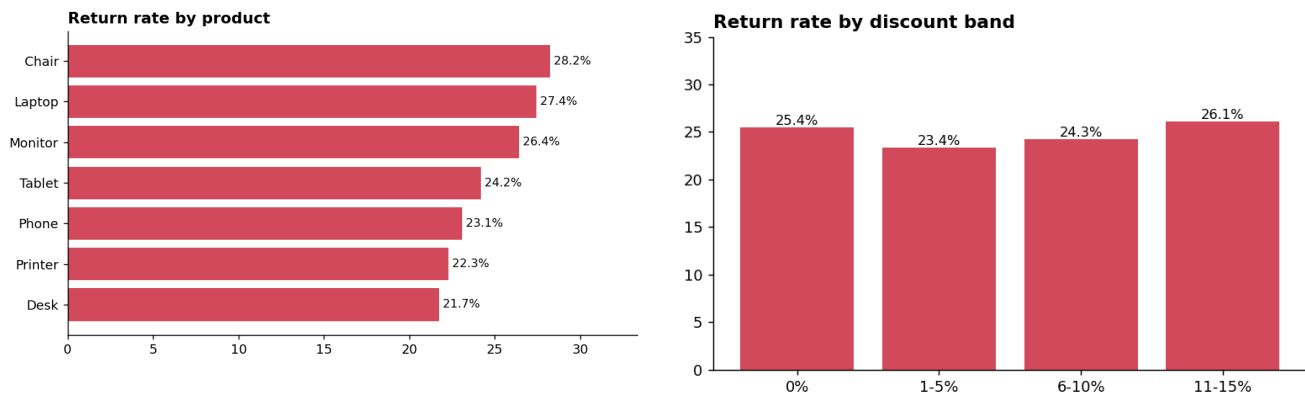
Customer type

CustomerType	Orders	Net sales	Units	Return rate	Avg delivery
Retail	744	\$2,195,526	7,680	25.4%	6.01 d
Wholesale	756	\$2,184,467	7,936	24.2%	6.08 d

Payment method

PaymentMethod	Orders	Net sales	Units	Return rate	Avg delivery
Online	323	\$971,115	3,423	23.8%	5.89 d
Cash	312	\$950,388	3,314	30.8%	5.93 d
Credit Card	299	\$866,483	3,080	21.7%	6.07 d
Gift Card	276	\$821,585	2,820	20.3%	6.29 d
Debit Card	290	\$770,421	2,979	26.9%	6.08 d

4.3 Returns and discounts



Figures 6-7. Return rate by product and by discount band.

Discount band return rates: 0% 25.4%, 1-5% 23.4%, 6-10% 24.3%, 11-15% 26.1%. The correlation between discount and return status is 0.007, effectively zero.

4.4 Order value and outliers

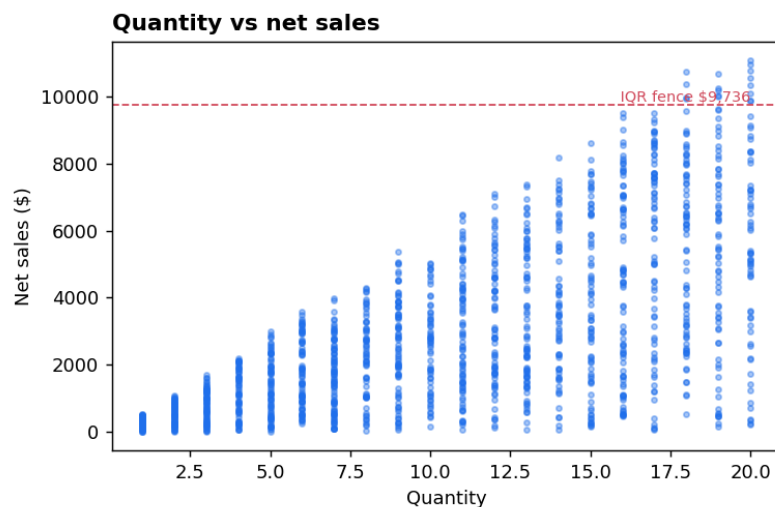


Figure 8. Quantity vs net sales. 17 orders sit above the IQR fence of \$9,736; these are review candidates, not errors. Mean \$2,920 vs median \$2,175 shows a right-skewed order value.

4.5 Matrices

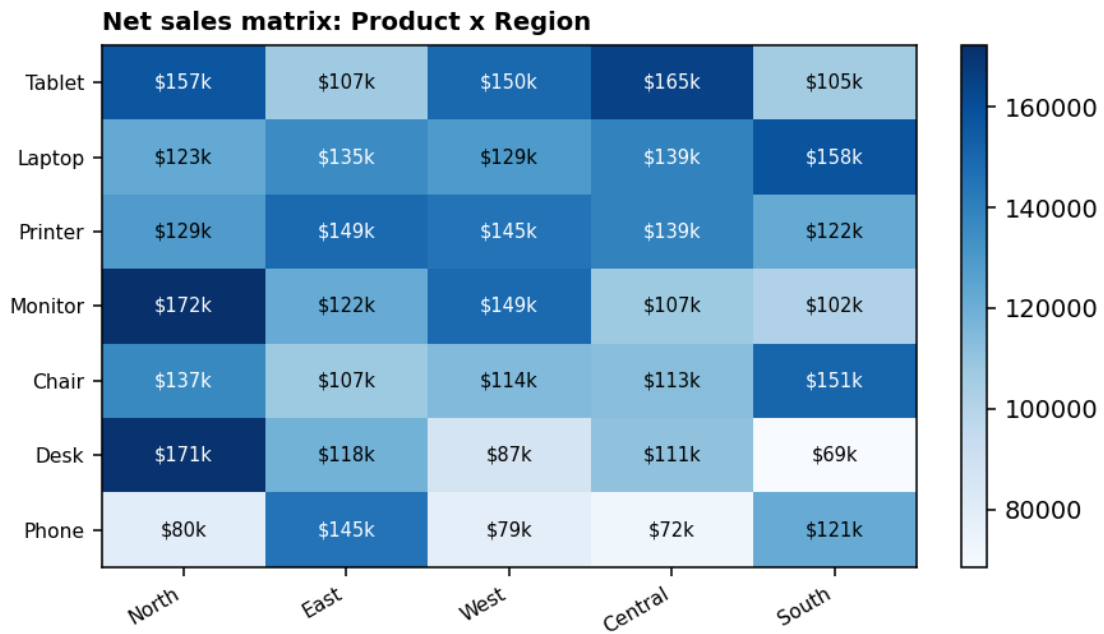


Figure 9. Net sales by Product x Region (equivalent to the dashboard matrix visual).

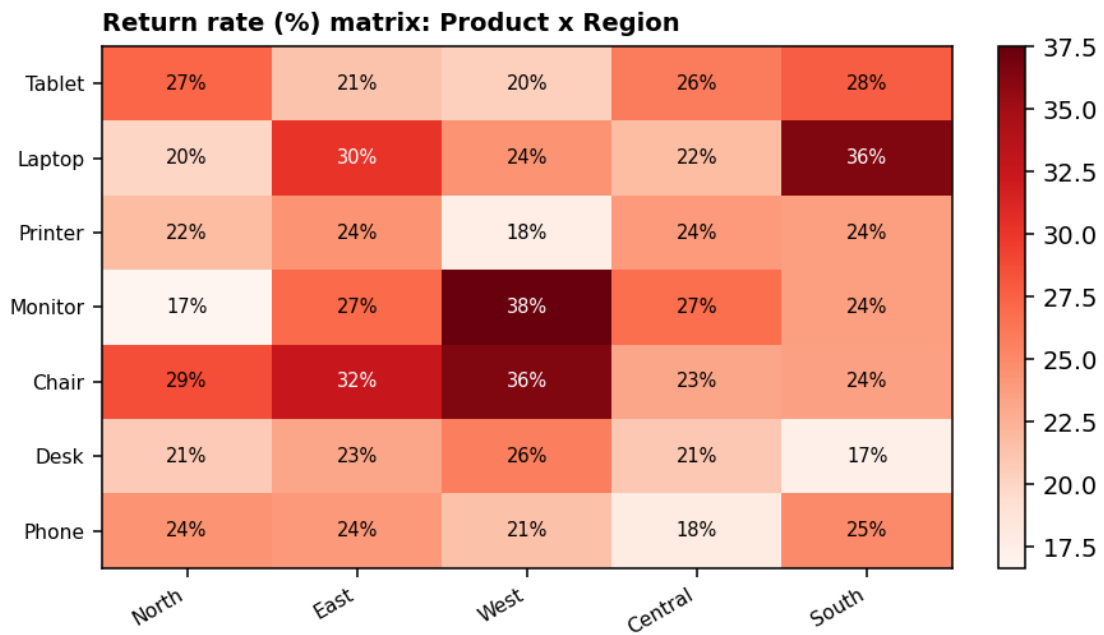


Figure 10. Return rate by Product x Region. Cells are small (about 25-50 orders), so read differences cautiously.

Product x Region net sales (table)

Product	North	East	West	Central	South	Total
Tablet	\$157,017	\$106,910	\$149,878	\$165,311	\$105,422	\$684,539
Laptop	\$122,644	\$135,461	\$129,397	\$139,220	\$157,696	\$684,417
Printer	\$128,877	\$149,018	\$145,227	\$139,175	\$122,091	\$684,387
Monitor	\$172,222	\$121,548	\$148,938	\$107,232	\$101,689	\$651,629
Chair	\$136,983	\$107,210	\$114,389	\$113,029	\$150,978	\$622,589
Desk	\$170,638	\$118,255	\$87,064	\$110,744	\$68,566	\$555,267
Phone	\$79,577	\$145,231	\$78,586	\$72,443	\$121,326	\$497,163
Total	\$967,958	\$883,634	\$853,479	\$847,154	\$827,768	\$4,379,992

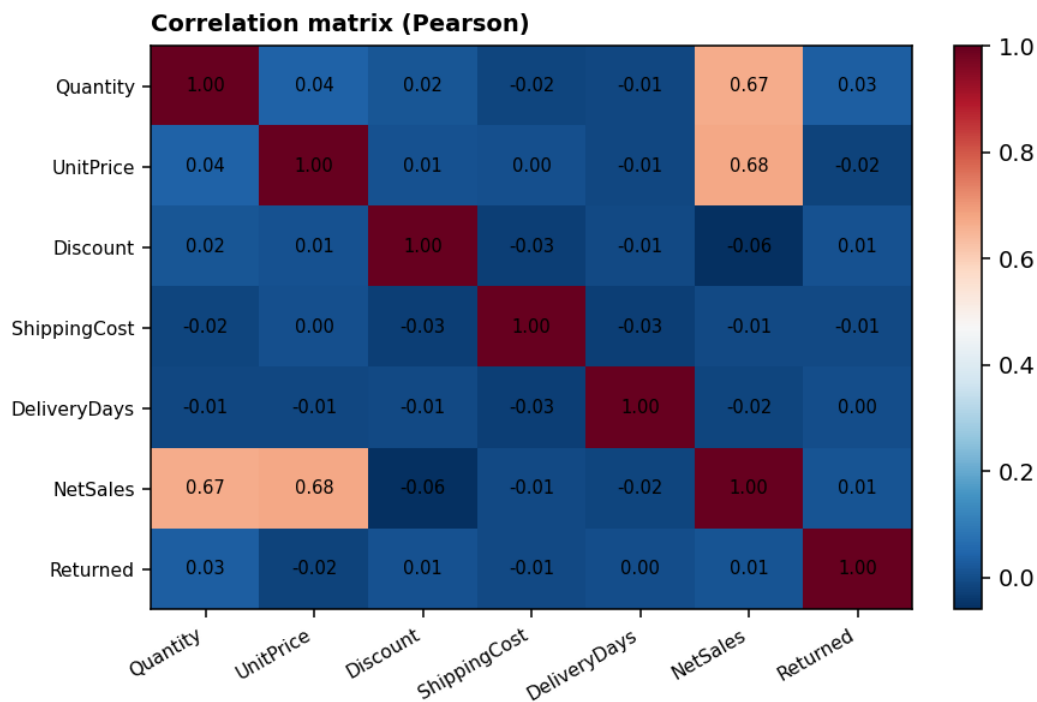


Figure 11. Correlation matrix. Quantity (0.67) and unit price (0.68) relate to net sales mechanically; delivery days (-0.02) and discount do not. Correlation is not causation.

5. Dashboard (Power BI)

Dashboard.pbix is a one-page interactive report built on the cleaned data, with a Year slicer that filters every visual. Its contents were read from the file's report layout:

Visual	Fields	Question answered
KPI card	Totals	Headline net sales / orders
Line chart	Year, Month	Trend and seasonality (Fig. 1)
Column chart	Region	Revenue by region (Fig. 2)
Bar + column charts	Product	Product ranking and mix (Fig. 3)
Bar chart	Salesperson	Individual performance (Fig. 4)
Column chart	Promotion	Campaign contribution (Fig. 5)
Bar chart	Discount	Sales at each discount level
Matrix	Product x Region	Where each product sells (Fig. 9)
Slicer	Year	Filters all visuals

The .pbix is a desktop file, so the dashboard's visuals are reproduced in this report as Figures 1-5 and 9, calculated from the same cleaned data. Open the file in Power BI Desktop for the interactive version.

6. Key business insights and recommendations

- 1. Revenue is broad-based.** North leads at \$967,958, but the gap to South is only \$140,190. Growth must come from lifting every region.
- 2. Three products carry the mix.** Tablet, Laptop, Printer each earn about \$684k. Phone (\$497,163) and Desk are weakest: candidates for promotion or range review.
- 3. Returns are the biggest leak.** 24.8% of orders are returned. Chair (28.2%) and Laptop (27.4%) are worst; East, West and South exceed 26%.
- 4. Discounts do not buy loyalty.** Discounts cost \$347,901 with no link to returns ($r = 0.007$). Test tighter discount caps.
- 5. Promotions.** FREESHIP drives the most net sales; SAVE10 has the highest return rate (26.0%).
- 6. Momentum is flat.** 2024 grew 4.4% over 2023, but first-half sales are level across all three years and the 2025 average order fell to \$2,701.
- 7. Sales team.** Bob (\$796,781) and Alice lead; Diana (\$676,568) is lowest.
- 8. Large orders need review.** 17 orders above \$9,736 can swing totals; verify pricing, quantity and promotions.

Recommended actions

1. Launch a returns root-cause review for Chair, Laptop and the East / West / South regions.
2. Set discount guardrails and A/B test 0-5% against 11-15% offers.
3. Investigate the Oct 2023 dip and plan around the March peak.
4. Share practices from top salespeople with the rest of the team.
5. Add cost and margin data so the next iteration measures profit, not only sales.

7. Limitations

2025 is a half year and should not be compared with full years. Correlations are not causal. Return rate is order-level, not unit-level. No profitability data is available, so shipping cost is not treated as profit. Findings describe this dataset only.